

OVERVIEW OF C & ITS DATATYPES

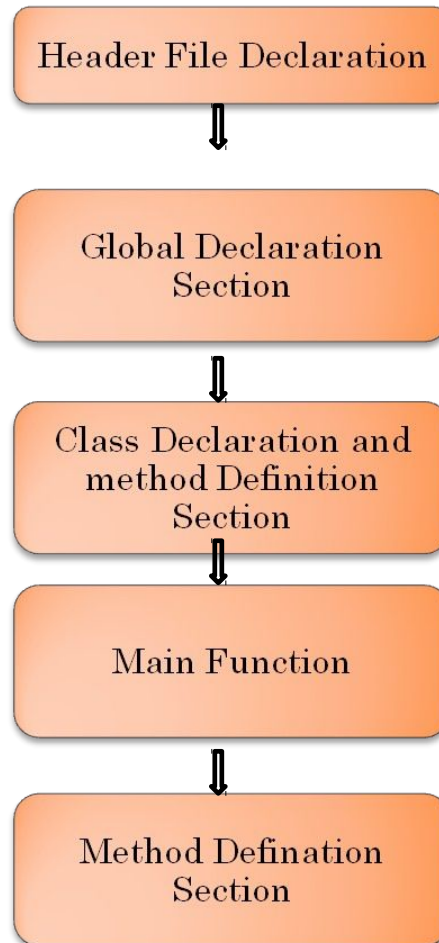
INTRODUCTION

- C is developed by Dennis Ritchie at Bell Labs in the mid 1970s.
- C is a high-level and general purpose programming language that is ideal for developing firmware or portable applications.
- It has been widely used both for operating systems and applications.
- Evolved from B, which evolved from BCPL

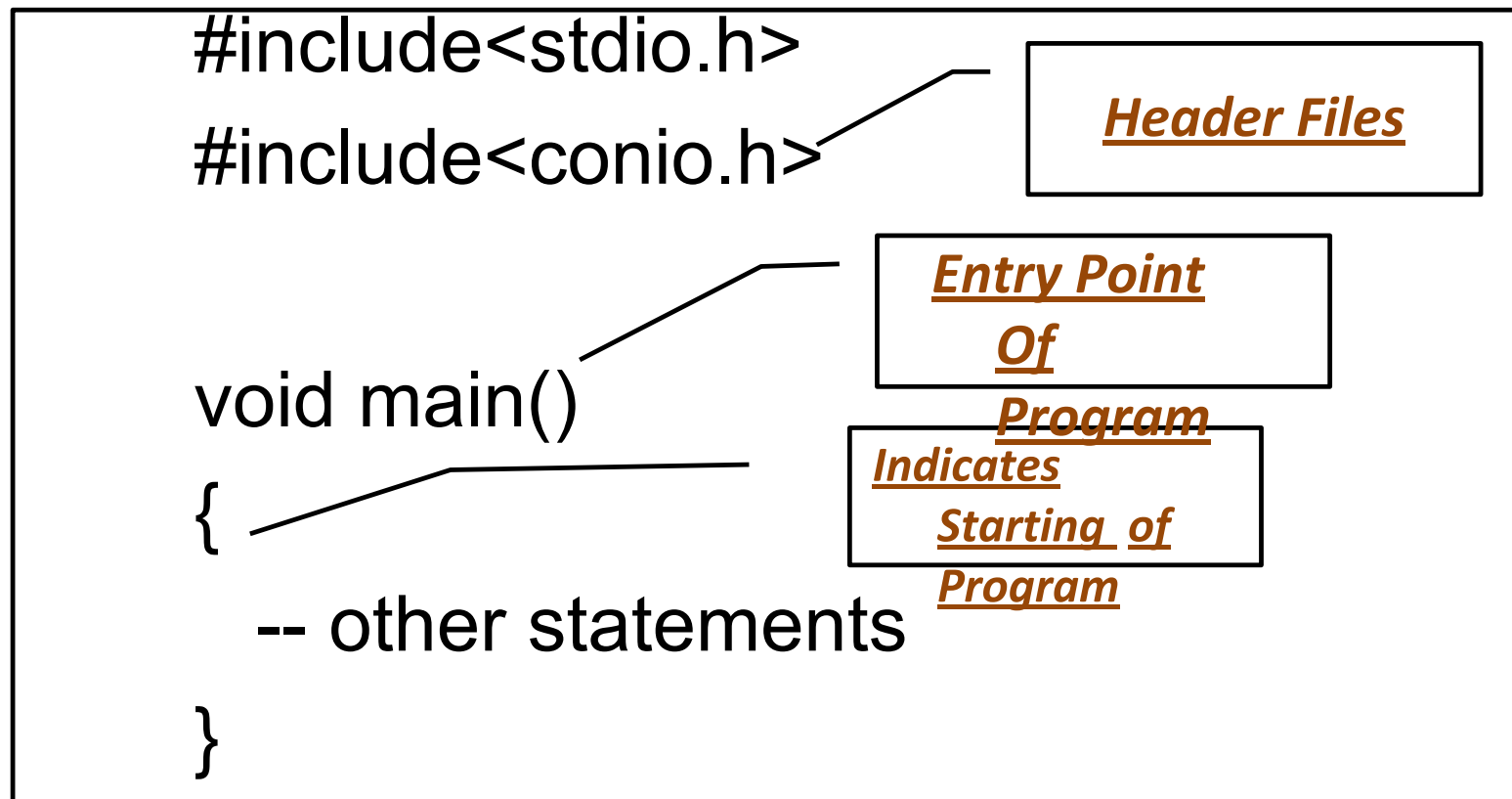
FEATURES OF C

- C provides features that are generally provided by the Lower level languages. C is Closely Related to Lower level Language such as “Assembly Language”.
- C Programs are **portable** i.e they can be run on any Compiler with little or no modification.
- Provides Wide variety of ‘Data Types’, ‘Functions’ and useful Control & Loop Statements
- It is more User friendly as compared to Previous languages.

Basic Structure of a C program



Basic Structure Of "C" Programs



Header files

- ✓ The files that are specified in the include section is called as Header File.
- ✓ These are precompiled files that has some functions defined in them.
- ✓ Header file is given an extension .h
- ✓ C Source file is given an extension .c

Main function

- ✓ This is the “*Entry Point*” of a program.
- ✓ When a file is executed, the start point is the main function.
- ✓ From main function the flow goes as per the programmers choice.
- ✓ There may or may not be other functions (user defined functions) written by user in a program.
- ✓ Main function is compulsory for any C program.

Running a 'C' Program

- Type a program.
- Save it.
- Compile the program – This will generate an executable file.
- Run the program .
- In different compilers, we have different option for compiling and running.

Example of “C” Program

```
#include <stdio.h>

main()

{
printf("Hello world\n");
}
```



Data Types in 'ansi c'

- **Data types** in c refer to an extensive system used for declaring variables or functions of different types.
- The type of a variable determines how much space it occupies in storage and how the bit pattern stored is interpreted.

Data Types in 'ansi c'

- ANSI C supports three classes of data types
 - Primary or Fundamental data types
 - ✓ int, float, double, char
 - Derived data types
 - ✓ Arrays, functions, pointers
 - User-defined data types
 - ✓ union, structure, enum, typedef

Data Types- Memory Requirements

Type	Size	Conversion characters
char	1 byte/ 8 bits	%c
int	2 bytes/ 16 bits	%d, %i, %o, %u, %x
float	4 bytes/ 32 bits	%f, %e, %g
double	8bytes/ 64 bits	%lf

THANK YOU